

Geospatial Project

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```
rm(list=ls())
library(readxl)      # read excel files
library(stringr)     # string operations
library(terra)       # raster operations
library(exactextractr) # exact_extract() for distances
library(gdistance)   # fast distance() formula for geo data
library(sf)          # vector operations
library(spData)      # geo data
library(tidyverse)   # NOTE: Conflict -- tidyr::extract() masks terra::extract()
library(spDataLarge) # large geo data
library(ggplot2)     # add labels to ggplot
library(tidyterra)   # visualize geospatial using tidy syntax
library(patchwork)   # patch tg ggplot objects
```

1 Abstract

Does the severity of earthquakes determine how much additional remittance income a municipality receives after the disaster happened? Unlike existing macro-level studies that rely on state data, this paper exploits more granular municipal level data to evaluate the impact of earthquake severity on post-disaster remittances. At this local level spatial spillovers between neighboring municipalities may violate SUTVA, introducing attenuation bias and artificially dampening the observed regional differences.

To investigate this, we construct a municipality-quarter panel of Mexican remittance inflows from Banco de México (2013 - 2025) (Banco de México 2025), merged with Peak Ground Acceleration (PGA) data from the USGS Earthquake Hazard Program (U.S. Geological Survey 2026). Using this dataset, we estimate a Two-Way Fixed Effects (TWFE) Difference-in-Differences model - **actually we probably need to use Callaway Sant’ Anna** :(-, where treatment intensity is defined as the average PGA experienced by each municipality.

2 Introduction

In 2024 alone, Mexico received more than 67 billion current US dollars in remittances, making it the second country in the world by nominal remittance inflows, behind India (World Bank Group 2026). When a shock hits the family in the municipality of origin, remittances act as an insurance mechanism, increasing flows for the following months Bettin, Jallow, and Zazzaro (2023). Mexico is one of the most seismically active countries in the world because of its location over three of the major tectonic plates (Rhea et al. 2011). Particularly, on 7 September 2017, a magnitude 8.2 earthquake struck the coast of Chiapas, followed after around two weeks by a magnitude 7.1 event near Puebla, the latter one causing the collapse of 44 buildings and severely damaging another 600 (Singh et al. 2018).

(Mohapatra, Joseph, and Ratha 2012) use cross-country data and find that remittances increase after natural disasters, especially where there is a large stock of migrant population. (Bettin, Jallow, and Zazzaro 2025) use monthly bilateral remittance data from Italy to 81 countries and estimate an immediate 2% increase following these type of events, reverting back to the normal flow after around one year. While precise, these papers aggregate at the country level and rely on a binary treatment definition. Our approach, in contrast, utilizes more granular data and a continuous measure of treatment. This allows us to test two competing outcomes: either we will identify a stronger, localized response effect in the hardest-hit municipalities, or we will find that network effects completely absorb these granular differences.

3 Data

3.1 Remittance Data

The outcome of our analysis is quarterly remittance inflows from Banco de México SIE table CE166 - *Ingresos por Remesas, distribución por municipio* -, covering all Mexican municipalities from Q1 2013 to Q4 2025.

```
# Read the excel skipping the metadata rows
remit_raw <- read_excel(
  "data/remittances/remittances.xlsx",
  skip = 8,
  col_types = "text"
)
```

We convert the dataset to long format and convert the date column in date format, adding columns for years and quarters to improve readability.

```
# Convert to long format
remit_long <- remit_raw %>%
  select(-`Information type`) %>% # Drop levels columns
  pivot_longer(
    cols      = -c(Title, Serie),
    names_to  = "quarter_label",
    values_to = "remittances_musd"
  ) %>%
  mutate(remittances_musd = as.numeric(remittances_musd))

# Convert quarter column to date format
remit_long <- remit_long %>%
  mutate(
    date      = as.Date(as.numeric(quarter_label), origin = "1899-12-30"),
    year      = as.integer(format(date, "%Y")),
    quarter   = ceiling(as.integer(format(date, "%m")) / 3)
  ) %>%
  select(-quarter_label) # drop the old quarter column
```

Being interested only in municipality level data, we drop the State aggregate rows. Moreover, to improve readability once more, we add the corresponding state column to the left of the municipality one.

```

# Tidy up municipality names
remit_long <- remit_long %>%
  # keep only remittances rows (drop state-level aggregates)
  filter(str_detect(Title, "Distribution by municipality")) %>%
  mutate(
    state = str_trim(str_extract(Title, "(?<=, )[^,]+(?:=, [^,]+$)")),
    municipality = str_trim(str_extract(Title, "[^,]+$"))
  ) %>%
  select(-Title) # drop the original Title column

# Drop unused columns and rearrange columns
remit_long <- remit_long %>%
  select(-date, -Serie) %>%
  select(state, municipality, year, quarter, remittances_musd)

```

3.2 Earthquake Data

4 Spillover Effects

4.1 Migration Networks

To get a measure of potential spillover effects, we can use a manually constructed weighting matrix of Mexican migration from a given Mexican municipality to each US state. We can then multiply this matrix by its transpose to get a municipality x municipality matrix to try and understand the connectedness of certain Mexican municipalities based on their shares US migration destinations. Theoretically, this means that certain Mexican municipalities that are not close in geographic distance can still propagate shocks through these migration networks.

```

M <- read_excel("C:/Users/T14/1BSE/8Thesis/1_network_estimation/2_migration_matrix_estim
muni_ids <- M[[2]] # store IDs
state_ids <- M[[1]]
muni_labels <- paste(state_ids, muni_ids, sep = "_") # generate unique label for each mu

M <- as.matrix(M[c(-1, -2)]) # drop names, convert to matrix
dim(M) # (2638 x 51) <==> (MX muni x US states) (District of Columbia gives us +1)

```

```
[1] 2638 51
```

```
# transpose matrix
M_prime <- t(M)

# MM' gives our (muni x muni) network
W_network <- M %*% M_prime
dim(W_network)
```

```
[1] 2638 2638
```

```
# place muni names back in
rownames(W_network) <- muni_labels
colnames(W_network) <- muni_labels

# municipality is not its own neighbor
diag(W_network) <- 0

# row normalize: add to one
W_network <- W_network/rowSums(W_network)

# sanity check
as.data.frame(W_network) %>%
  slice(1:4) %>%
  select(1:2)
```

	Aguascalientes_Aguascalientes
Aguascalientes_Aguascalientes	0.000000000
Aguascalientes_Asientos	0.005173659
Aguascalientes_Calvillo	0.005273214
Aguascalientes_Cosio	0.005130624

	Aguascalientes_Asientos
Aguascalientes_Aguascalientes	0.0004038910
Aguascalientes_Asientos	0.0000000000
Aguascalientes_Calvillo	0.0004379844
Aguascalientes_Cosio	0.0003668245

```
# largest values in a row = strongest network neighbors
# sort(W_network["Michoacan_Coeneo",], decreasing = TRUE)[1:5]
```

i Note

Note that we row-normalize our final matrix here; what is important is that this final product is not a symmetric matrix. The interpretation is that for a given row i (mu-

municipality), we get the probability distribution across all other municipalities ($\forall_{i \neq j}$). We must include it in this form in order to properly use it as a variable in our regression later on.

This spillover matrix serves to approximately measure the “connectedness” between every pair of municipalities in Mexico, based on their historical migration patterns to the US. The idea is that if an earthquake hits Asientos in Aguascalientes, the migration community in (say) Texas that comes from this municipality will send more money home. But if that same community in Texas also has a large network of migrants from Coeneo in Michoacan, they might discuss the earthquake; consequently, migrants from Coeneo may send additional remittances back home as feel a heightened sense of their own family’s vulnerability; in other words, they anticipate risk associated with disaster in another municipality.

4.2 Geographic Distance

Furthermore, we may suspect that neighboring municipalities in Mexico would contribute to this spillover effect. Theoretically, if an earthquake hits a municipality hard enough they might rely on aid from neighboring towns as local residents may be out of work and cannot afford to survive after such a disaster. Thus, we can create another matrix, this time using a simple weighting matrix based on geographic distance to the affected municipalit(ies).

To do so, we can simply use our previous weighting matrix and initialize the matrix by assigning zeros to all (municipality x municipality) pairs.

```
empty_network = W_network * 0 # init empty matrix
```

5 Regression: Callaway Sant-Anna

parallel trends ass

what regressors we include in the model

Banco de México. 2025. “Revenues by Workers’ Remittances, by Municipality (CE166).” <https://www.banxico.org.mx/SieInternet/consultarDirectorioInternetAction.do?sector=1&accion=consultarCuadro&idCuadro=CE166&locale=en>.

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